



WELCOME

Dataset Preview

	A	B	C	D	E	F	G	H	I	J	K	L
1	state_code	LGD_ST_C	state_name	district_code	LGD_Dist	district_name	total_no_of_job_cards_issued	total_no_of_workers	total_no_of_active_job_cards	total_no_of_active_workers	sc_worker	st_worker
2	1	35	ANDAMAN AND NICOBAR	101	602	SOUTH ANDAMAN	13979	17950	2466	3630	0	0
3	1	35	ANDAMAN AND NICOBAR	102	603	NICOBARS	6101	9739	943	1782	0	1414
4	1	35	ANDAMAN AND NICOBAR	103	632	NORTH AND MIDDLE ANDAMAN	18808	30347	4297	8107	0	0
5	2	28	ANDHRA PRADESH	201	519	SRIKAKULAM	444211	757458	381119	673422	60008	40167
6	2	28	ANDHRA PRADESH	202	521	VIZIANAGARAM	389011	694851	345963	634267	73287	17807
7	2	28	ANDHRA PRADESH	203	520	VISAKHAPATANAM	36975	57160	31834	50922	4341	487
8	2	28	ANDHRA PRADESH	204	505	EAST GODAVARI	170314	262643	126283	201588	72389	3338
9	2	28	ANDHRA PRADESH	205	523	WEST GODAVARI	187779	309725	137651	237911	64682	1907
10	2	28	ANDHRA PRADESH	206	510	KRISHNA	223428	395726	182585	337039	122418	5823
11	2	28	ANDHRA PRADESH	207	506	GUNTUR	138293	232051	91093	159704	70922	5697
12	2	28	ANDHRA PRADESH	208	517	PRAKASAM	457793	849764	379475	731972	200893	22423
13	2	28	ANDHRA PRADESH	209	515	NELLORE	361210	613300	273978	481109	139165	30093
14	2	28	ANDHRA PRADESH	210	503	CHITTOOR	276459	496122	210210	389424	110594	13554
15	2	28	ANDHRA PRADESH	211	504	Y.S.R	247499	456401	189045	355096	78707	5404
16	2	28	ANDHRA PRADESH	212	502	ANANTAPUR	326015	606925	259153	489307	95634	17984
17	2	28	ANDHRA PRADESH	213	511	KURNOOL	346503	662226	267367	526725	109088	7885
18	2	28	ANDHRA PRADESH	214	745	ALLURI SITHARAMA RAJU	258062	495340	234344	460494	9930	421541
19	2	28	ANDHRA PRADESH	215	744	ANAKAPALLI	292742	462803	238892	393652	35062	16949
20	2	28	ANDHRA PRADESH	216	747	KONASEEMA	237270	367270	154370	256653	118173	2386
21	2	28	ANDHRA PRADESH	217	748	ELURU	384457	695452	320722	598697	205762	54254
22	2	28	ANDHRA PRADESH	218	749	NTR	217592	393982	178690	331623	124485	24236
23	2	28	ANDHRA PRADESH	219	750	BAPATLA	275359	487459	216098	402451	140396	15413
24	2	28	ANDHRA PRADESH	220	751	PALNADU	354122	612117	275288	496151	123600	45472

NUMBER OF ROWS - 740
NUMBER OF COLUMNS - 34

Name Of Columns

1. state_code
2. LGD_ST_Code
3. state_name
4. district_code
5. LGD_Dist_Code
6. district_name
7. total_no_of_job_cards_issued
8. total_no_of_workers
9. total_no_of_active_job_cards
10. total_no_of_active_workers
11. sc_worker_percentage_against_active_workers
12. st_worker_percentage_against_active_workers
13. approved_labour_budget
14. persondays_of_central_liability_so_far
15. sc_persondays_percentage_of_total
16. st_persondays_percentage_of_total
17. women_persondays_percentage_of_total
18. average_days_employment_per_household
19. average_wage_rate_per_day_per_person
20. total_no_of_HHs_completed_100_days
21. total_households_worked
22. total_individuals_worked
23. differently_abled_persons_worked
24. no_of_GPs_with_NIL_expenditure
25. total_no_of_works_taken_up_new_and_spillover
26. no_of_ongoing_works
27. no_of_completed_works
28. percentage_of_NRM_expenditure_in_MWC_blocks
29. percentage_of_category_B_works
30. percentage_expenditure_on_agriculture_and_allied_works
31. total_expenditure
32. wages
33. material_and_skilled_wages
34. total_admin_expenditure

MGNREGA: A Data - Based Comparative Study of Indian States



MGNREGA: A Data - Based Comparative Study of Indian States



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In Collaboration with
Institute of Chemical Technology, Mumbai
November 2025



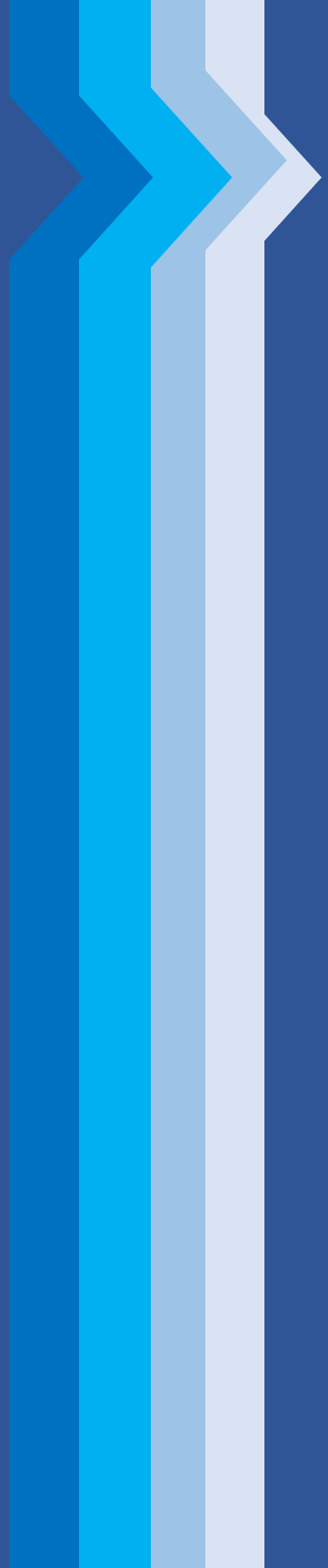
INTRODUCTION.

- MGNREGA is a rural employment and social welfare program launched in 2005.
- It guarantees 100 days of wage work each year to rural households.
- Aims to reduce poverty, create rural assets, and empower communities.
- Implemented by Gram Panchayats across India.
- Reflects Gandhiji's vision of self-reliance and dignity of labour.



Rights of Workers.

- i. Demand Work: Employment within 15 days of request.
- ii. Job Card: With member details and work record.
- iii. Minimum Wages: Equal to or above state rates.
- iv. Worksite Facilities: Water, shade, first-aid.
- v. Timely Payment: Wages within 15 days or compensation.
- vi. Right to Information: Access to records under RTI.
- vii. Grievance Redressal: Report wage or work issues



1991

Concept of a
statutory
employment
guarantee
first proposed.

Evolution Of MGNREGA

1991

Concept of a statutory employment guarantee first proposed.

2005

National Rural Employment Guarantee Act (NREGA) passed by Parliament.

2006

Scheme launched in 200 of the most backward districts (Feb 2nd).

2008

Scheme expanded to cover all rural districts of India (April 1st).

2009

Renamed to "MGNREGA" in honor of Mahatma Gandhi (Oct 2nd).

Role of Gram Panchayats under MGNREGA.

1. Registration of Households:

- Identify and register rural households willing to work.
- Issue Job Cards to eligible families.

2. Planning and Approval:

- Prepare Annual Work Plan & Labour Budget (AWP&LB).
- Approve works in Gram Sabha meetings for public participation.

3. Implementation of Works:

- Execute and supervise projects like roads, ponds, and water conservation.
- Maintain worksite discipline and safety.

4. Maintenance of Records:

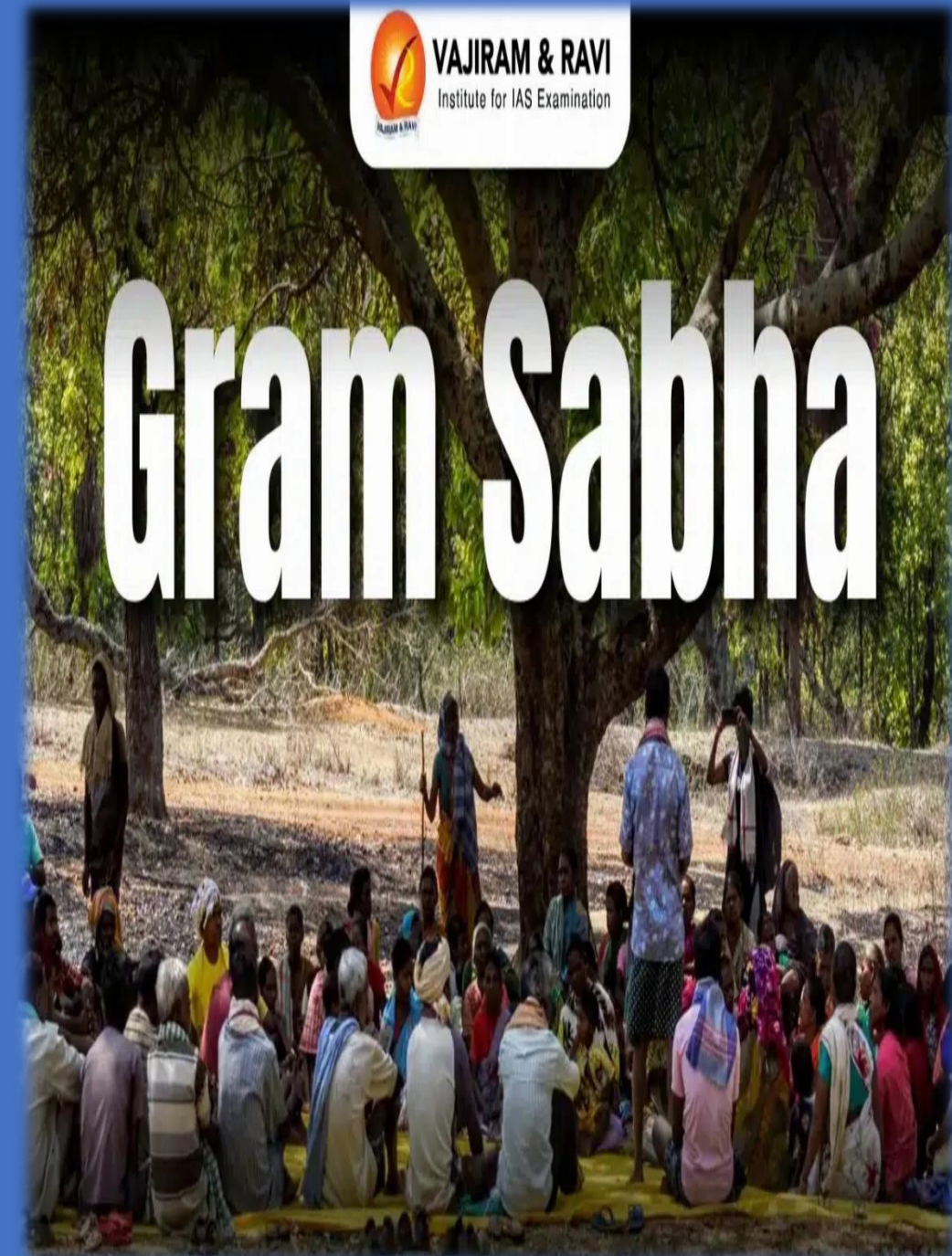
- Keep attendance, payment, and material registers.
- Upload progress and expenditure on the MIS portal.

5. Social Audits and Transparency:

- Conduct bi-annual social audits.
- Promote community monitoring to prevent corruption.

6. Coordination with Higher Authorities:

- Work with BDOs and District Programme Coordinators for implementation.



Guaranteed Employment

Provide 100 days of guaranteed wage employment to rural households to secure livelihoods and reduce poverty.

G

Social Empowerment

Promote social inclusion by ensuring active participation of women, scheduled castes (scs), and scheduled tribes (sts).

S

Rural Development

Create durable assets like water conservation structures, roads, and irrigation facilities to boost rural infrastructure

R

Local Governance

Strengthen the role of Gram Panchayats and local institutions in planning, implementing, and monitoring projects.

L

Implementation Mechanism Of MGNREGA

CENTRAL

01

Central Level

- **Managed by Ministry of Rural Development (MoRD)**
- **Handles policy, funding, and monitoring**

Implementation Mechanism Of MGNREGA

STATES

02

State Level

- Oversees implementation and fund transfer
- Uses MGNREGA MIS portal for transparency

Implementation Mechanism Of MGNREGA

DISTRICT

03

District Level

- **District Collector (DPC) supervises all blocks and ensures timely project approval**

Implementation Mechanism Of MGNREGA

BLOCK

04

Block Level

- **Programme Officer (PO) coordinates with Gram Panchayats and monitors work progress.**

Implementation Mechanism Of MGNREGA

GRAM

05

Gram Panchayat Level

- **Registers workers, issues Job Cards, and plans local projects.**
- **Ensures local participation and accountability.**

Financial Provisions & Funding Pattern

Financial Provisions & Funding Pattern



01

Central-State Cost Sharing

- Registers workers, issues Job Cards, and plans local projects.
- Ensures local participation and accountability.



02

Budget Allocation

- **Annual funds based on labour demand, performance, and utilization.**
- **Released in installments after progress and utilization reports.**



03

Fund Flow Mechanism

- **Funds transferred from Centre → SEGF
→ District/Block accounts.**
- **Electronic transfers ensure
transparency and reduce delays.**



04

Wage Payments

- **Wages credited via Direct Benefit Transfer (DBT) to workers accounts.**
- **Must be paid within 15 days, with compensation for delays.**

05

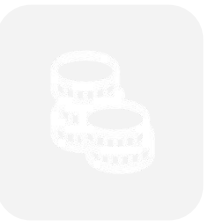
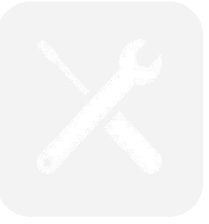
Monitoring & Accountability

- Financial data managed through MGNREGA MIS portal (publicly accessible).
- Social audits and independent reviews ensure transparency and proper use of funds.

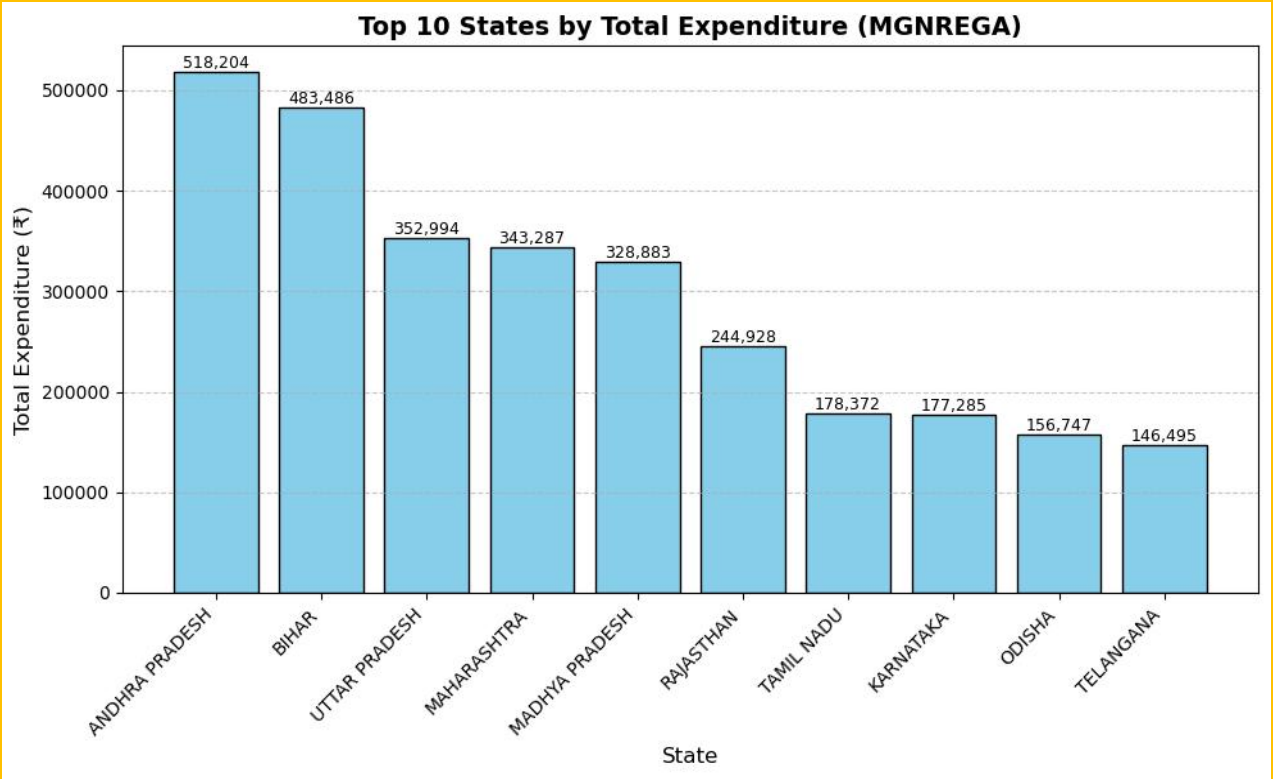




Graphical Representation



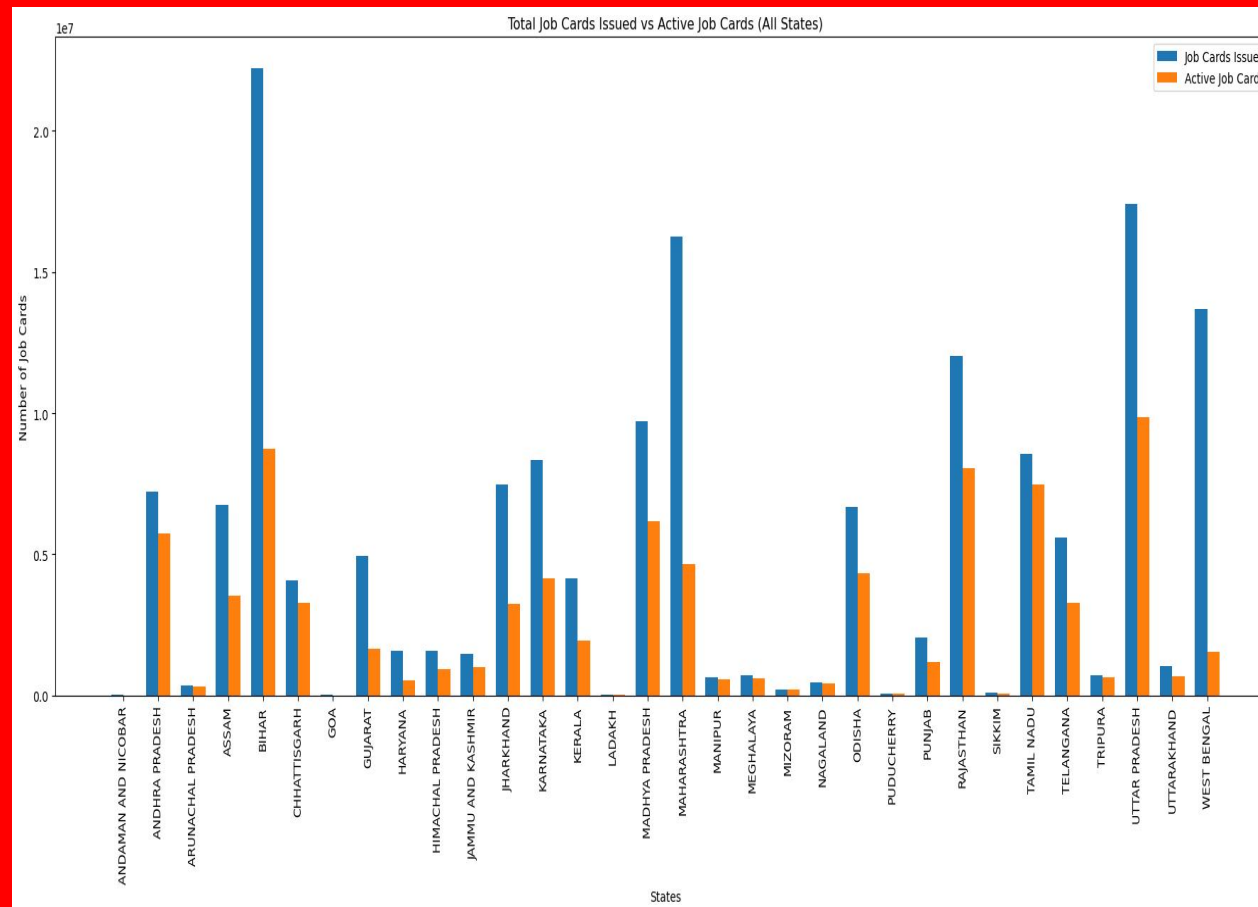
Total Expenditure (Top 10)



- 1. Andhra Pradesh spent the highest under MGNREGA.
- 2. Bihar ranks second with strong employment demand.
- 3. UP, Maharashtra & MP take a major share due to large rural populations.
- 4. Other states show consistent fund utilisation.
- 5. Total expenditure $\approx$ ₹29.31 lakh crore across top 10 states.

Graphical Representation

Total Job Cards Issued vs Total Active Job Cards



High States

- Bihar, UP, Maharashtra, West Bengal, Rajasthan
- Very high job cards; active cards vary but major gap in some states.

Moderate States

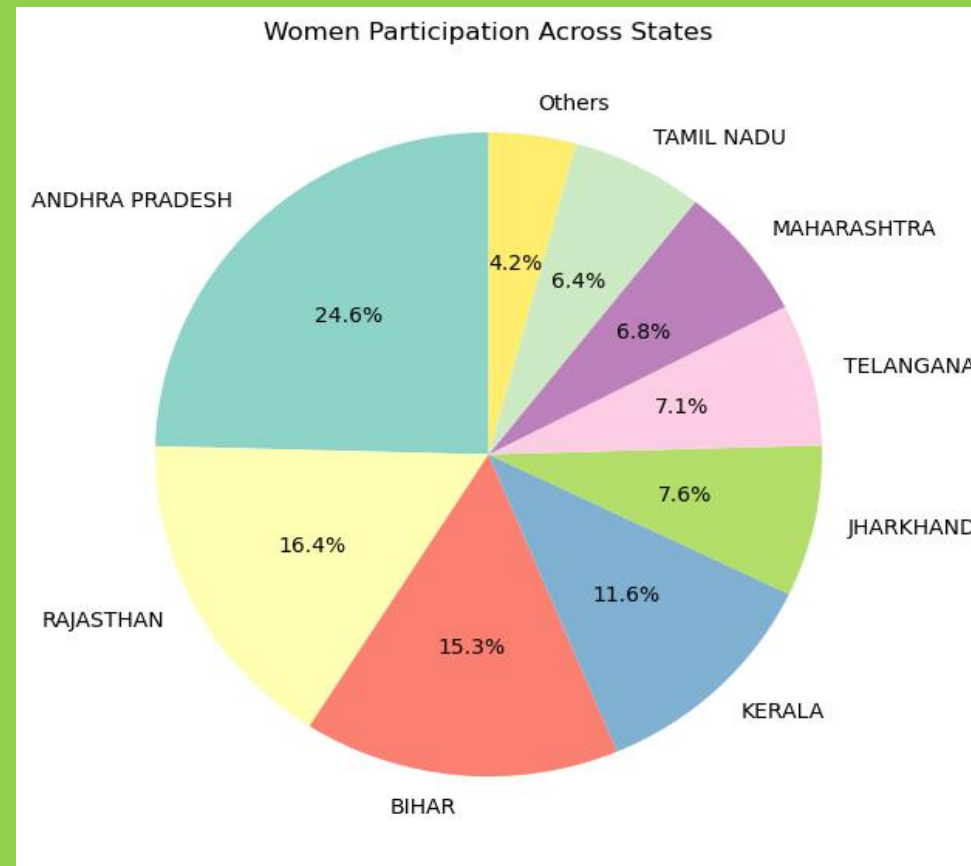
- AP, Assam, Jharkhand, Karnataka, MP, Odisha, Tamil Nadu, Telangana
- Moderate job cards; medium to good active participation.

Low States

- Goa, Sikkim, Mizoram, Nagaland, Meghalaya, Arunachal Pradesh
- Low job cards due to small population; low activity levels.

Graphical Representation

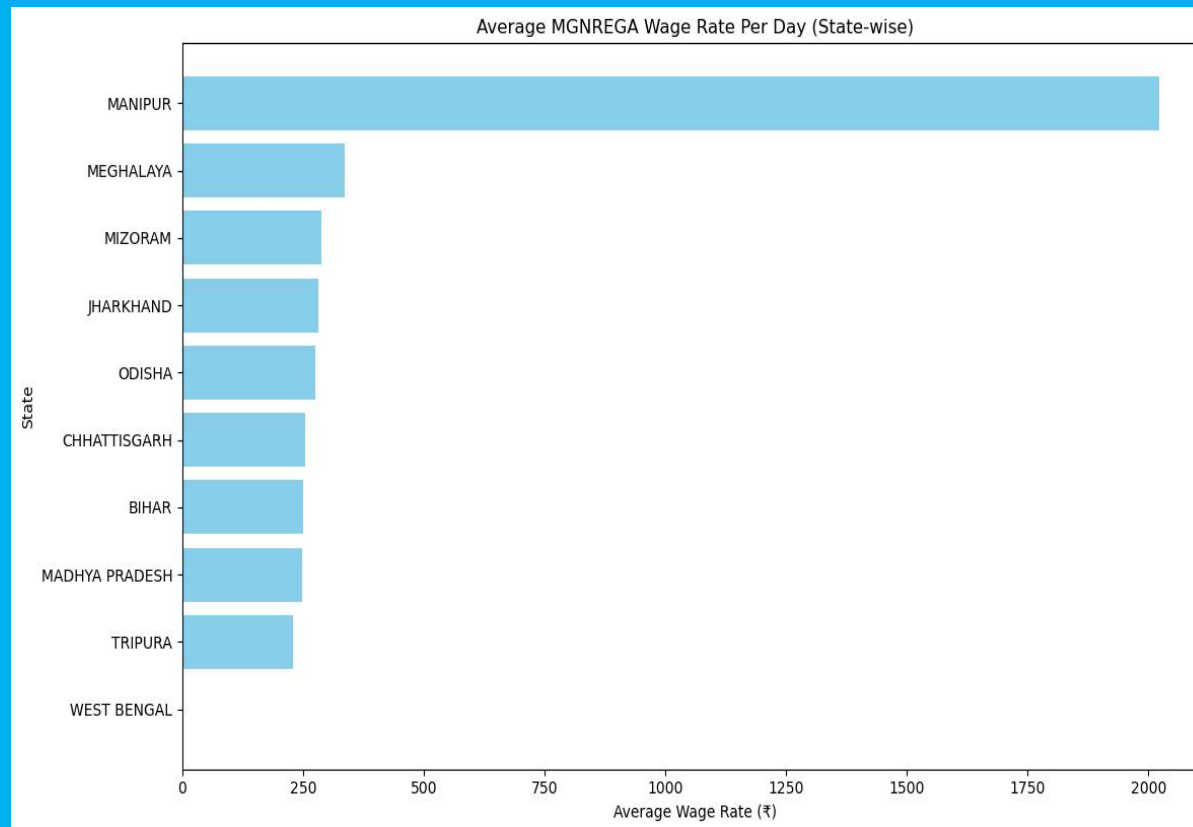
Women Participation%



1. Andhra Pradesh shows the highest women participation.
2. Rajasthan & Bihar also have significant involvement.
3. Kerala, Jharkhand & Telangana show moderate shares.
4. Other states contribute comparatively lower.
5. Highlights strong female involvement in MGNREGA.

Graphical Representation

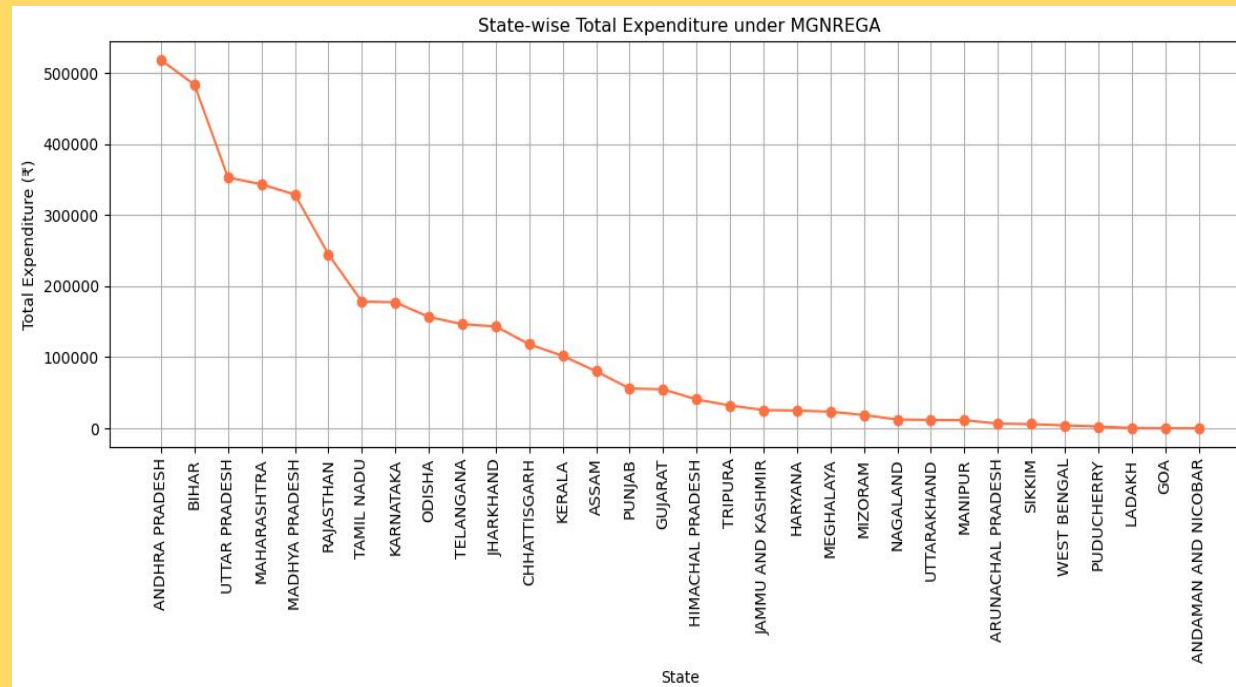
State-wise Average Wages



- Manipur shows extremely high wages (₹2000+), standing out as an outlier.
- Other states have much lower wages (₹200–300), showing a wide disparity.
- Northeastern states (Meghalaya, Mizoram) offer moderately higher wages than central/eastern states.
- Bihar, Jharkhand, MP, Chhattisgarh have among the lowest wage rates.
- Chart highlights significant state-wise variation in MGNREGA wages.

Graphical Representation

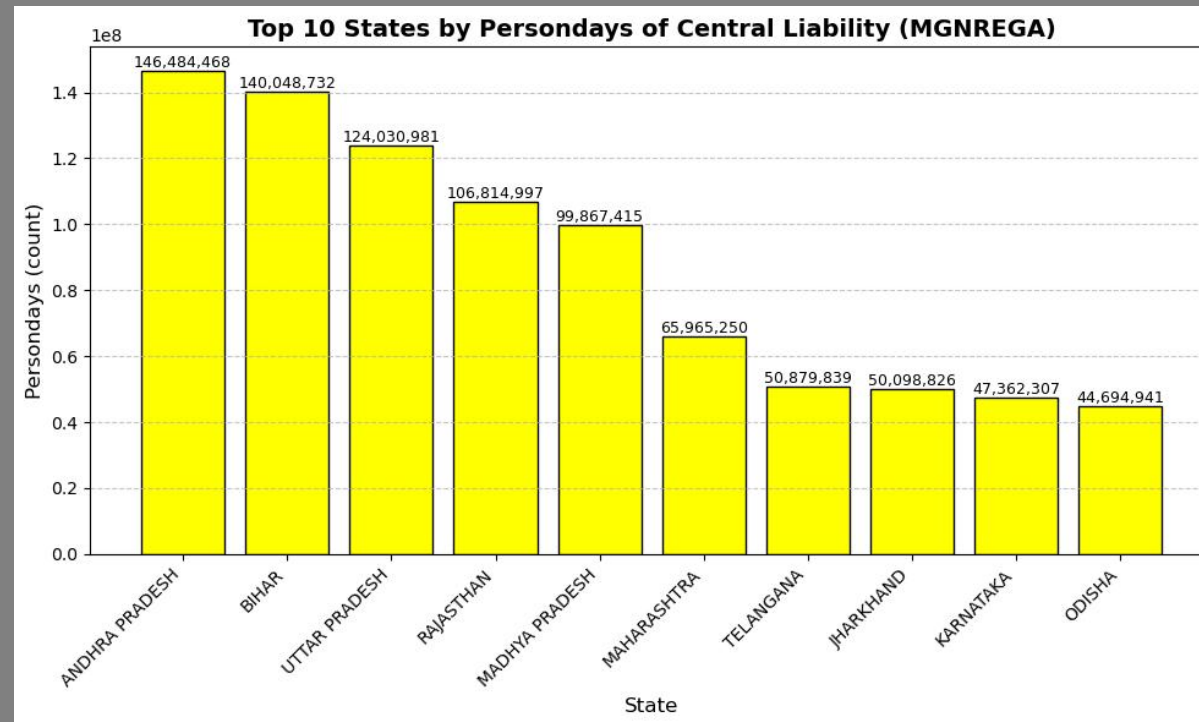
State-wise Total Expenditure



- Andhra Pradesh shows the highest MGNREGA spending.
- Bihar and Uttar Pradesh also have high expenditure levels.
- Maharashtra, Madhya Pradesh, and Rajasthan show moderate spending.
- Southern states (TN, Karnataka, Telangana) show steady expenditure.
- Northeastern states record low expenditure.
- UTs and small states like Goa, Ladakh, and A&N have the lowest spending.
- Overall, huge variation seen among states' expenditure levels.

Graphical Representation

Person-days generated (Top 10)



- Andhra Pradesh leads with 1,464.84 lakh persondays.
- Bihar follows with 1,400.49 lakh, showing high rural demand.
- UP, Rajasthan & MP generate 900+ lakh, indicating wide rural reach.
- Maharashtra & Telangana: moderate levels (500–660 lakh).
- Jharkhand, Karnataka & Odisha: 450–500 lakh, still top performers.

Statistical Analysis (R Programming)

Overview

- Analysis performed using R programming to study MGNREGA trends.
- Used statistical tests to examine:
 1. Expenditure
 2. Employment (person-days)
 3. Participation
 4. Wage levels

Methods Used

- Correlation Analysis - Measures the relationship between total expenditure and person-days generated.
- ANOVA (Analysis of Variance) - Tests whether employment levels differ significantly across states.

Purpose of Analysis

- To measure effectiveness of spending
- To compare state-wise performance
- To understand the influence of gender participation
- To identify wage differences

Statistical Analysis (R Programming)

5.1 Correlation Between Expenditure and Employment

Objective:

To study whether total expenditure under MGNREGA is positively associated with the employment generated (in person-days).

Method:

The Pearson correlation coefficient (r) measures the strength and direction of this linear relationship.

Formula:
$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$$

R Code:

```
> #Correlation analysis
> correlation <- cor(df$total_expenditure, df$average_days_employment_per_household, method = "pearson")
> correlation
[1] 0.453986
```

The correlation value of 0.453986 shows a moderate positive relationship between expenditure and employment. This means higher government spending generally leads to more employment under MGNREGA, though other factors like administration and regional conditions also affect outcomes.

5.2 Analysis of Variance (ANOVA)

Objective:

To test if mean employment generated differs significantly among states categorized by different expenditure levels.

Hypothesis:

- H_0 : Mean employment across expenditure levels is equal.
- H_1 : At least one group differs.

R Code:

```
> # Categorize Total Expenditure into 3 levels: Low, Medium, High
> df$Expenditure_Level <- cut(df$total_expenditure,
+                             breaks = 3,
+                             labels = c("Low", "Medium", "High"))
>
> # Perform ANOVA: Check if persondays generated differ by expenditure level
> anova_model <- aov(persondays_of_central_liability_so_far ~ Expenditure_Level, data = df)
>
> # Display the summary of the ANOVA model
> summary(anova_model)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Expenditure_Level	2	1.150e+15	5.751e+14	439.7	<2e-16 ***
Residuals	736	9.627e+14	1.308e+12		

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Case Studies / Examples



Rajasthan – Water Conservation Success

- Built check dams & ponds → revived groundwater, improved irrigation.
- Created jobs & ensured water sustainability.

Case Studies / Examples



Bihar & Madhya Pradesh – Rural Infrastructure

- Built rural roads & land works → better connectivity, higher farm productivity.
- Boosted employment & rural growth.

Case Studies / Examples

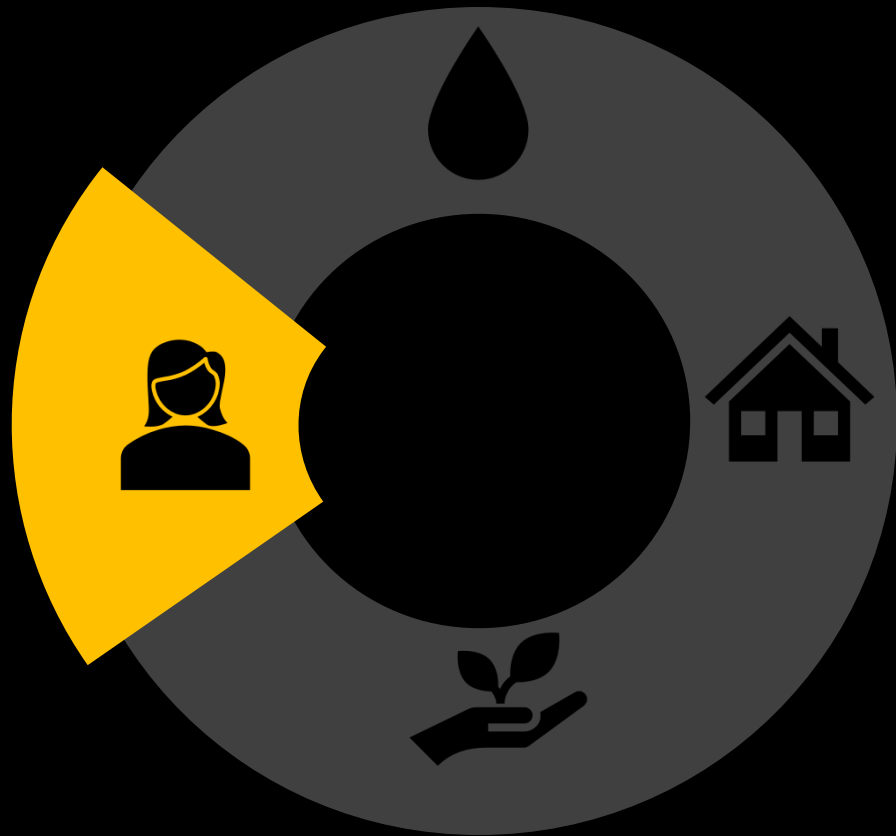


Odisha – Agricultural Empowerment

- Repaired canals & reclaimed land → higher yields, less migration.
- Improved food security & self-sufficiency.

Case Studies / Examples

Kerala – Women Empowerment (Kudumbashree)



- Kudumbashree groups managed afforestation & waste works.
- Women gained leadership & steady income.



Impact: Social, Economic & Environmental

Social Impact

- Reduced poverty and migration.
- Empowered women through income and decision-making.
- Improved inclusion of SC/ST/OBC communities.
- Increased participation in local governance.

Economic Impact

- Boosted rural purchasing power and local economies.
- Created durable assets (roads, irrigation).
- Supported small businesses and improved agriculture.

Environmental Impact

- Water conservation improved groundwater levels.
- Afforestation increased green cover.
- Soil and watershed management boosted agricultural sustainability.



Challenges & Limitations

Delayed Wage Payments

Complex fund-flow systems and slow approvals cause delays. Workers depend on timely wages for survival; delays reduce participation and encourage migration.

Leakages & Corruption

Fake job cards, inflated attendance, ghost beneficiaries, and false muster rolls divert funds. Weak monitoring at Gram Panchayat level enables manipulation.

Lack of Awareness

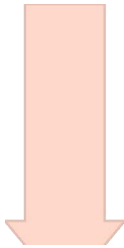
Many rural households unaware of rights: 100 days guarantee, equal wages, unemployment allowance. Poor communication limits participation of women and marginalized groups.

Data & Administrative Issues

Delayed data reporting, mismatch between records and ground reality, weak digital infrastructure, and administrative overload reduce transparency and efficiency.

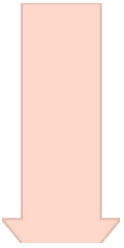


Solutions & Way Forward



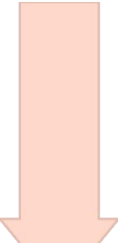
Strengthen Digital Systems

Expand Ne-FMS, Aadhaar-DBT, and real-time monitoring dashboards to ensure timely wage payments and reduce leakages.



Enhance Transparency

Implement biometric attendance, geo-tagging of assets, and social audits. Promote community oversight through online portals and public disclosure.



Improve Awareness

Strengthen grassroots campaigns through local languages, community radio, and village volunteers. Collaborate with NGOs and self-help groups.



Build Capacity

Provide regular training to local officials, strengthen coordination between central and local authorities, and ensure strict accountability mechanisms.

Conclusion: MGNREGA's Transformative Role

MGNREGA remains India's most significant rural welfare program, providing livelihood security to millions while creating durable assets and promoting social equity. With 87.62 crore person-days generated and strong women participation, it has reduced migration and empowered marginalized communities. Environmental benefits through water conservation and afforestation align employment with sustainability. Despite challenges in wage payments, corruption, and awareness, technological improvements and stronger governance can enhance effectiveness. MGNREGA embodies inclusive growth—a cornerstone in India's journey toward equitable rural development and social justice.



THANK YOU

Total Expenditure (Top10)

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_excel("C:/Users/Admin/Downloads/MGNREGA_dataset_AtAGlance18.xlsx")
df.columns = df.columns.str.strip().str.lower()
df = df.rename(columns={'state': 'state_name', 'expenditure': 'total_expenditure'})
df['total_expenditure'] = pd.to_numeric(df['total_expenditure'], errors='coerce')

top10 = df.groupby('state_name')['total_expenditure'].sum().nlargest(10)

plt.figure(figsize=(10, 6))
bars = plt.bar(top10.index, top10.values, color='skyblue', edgecolor='black')
for bar in bars:
    plt.text(bar.get_x() + bar.get_width()/2, bar.get_height(),
             f'{bar.get_height():,.0f}', ha='center', va='bottom', fontsize=9)

plt.title("Top 10 States by Total Expenditure (MGNREGA)", fontsize=14, weight='bold')
plt.xlabel("State", fontsize=12)
plt.ylabel("Total Expenditure (₹)", fontsize=12)
plt.xticks(rotation=45, ha='right')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```

Number of Households Worked vs Completed 100 Days

```
import matplotlib.pyplot as plt
import pandas as pd

df=pd.read_excel("C:/Users/Admin/Downloads/MGNREGA_dataset_AtAGlance18.xlsx")
data=df.groupby("state_name")[["total_households_worked", "total_no_of_HHs_completed_100_days"]].sum()

data.plot(kind='bar', figsize=(12,6), color=['yellow', 'Black'])
plt.title("Number of Households Worked vs Completed 100 Days")
plt.xlabel("State")
plt.ylabel("Number of Households")
plt.legend(["Worked", "Completed 100 Days"])
plt.tight_layout()
plt.show()
```


Women Participation%

```
women_participation = df.groupby("state_name")["women_persondays_percentage_of_total"].mean()

top8 = women_participation.sort_values(ascending=False)[:8]
others = pd.Series({"Others": women_participation[8:].mean()})
pie_data = pd.concat([top8, others])

plt.figure(figsize=(7,7))
pie_data.plot(kind='pie', autopct='%1.1f%%', startangle=90, colormap='Set3')
plt.title("Women Participation Across States")
plt.ylabel("")
plt.show()
```


State-wise Average Wages

```
import pandas as pd
import matplotlib.pyplot as plt

file_path = "C:/Users/Admin/Downloads/MGNREGA_dataset_AtAGlance18.xlsx"
df = pd.read_excel(file_path, sheet_name="State-Wise")

# Group by state and calculate average wage
avg_wage_per_state = (
    df.groupby("state_name")["average_wage_rate_per_day_per_person"]
      .mean()
      .reset_index()
)

# Sort descending (highest wage first)
avg_wage_per_state = avg_wage_per_state.sort_values(
    by="average_wage_rate_per_day_per_person",
    ascending=False
)

# Display table
print(avg_wage_per_state)

# Plot horizontal bar chart
plt.figure(figsize=(12, 7))
plt.barh(avg_wage_per_state["state_name"], avg_wage_per_state["average_wage_rate_per_day_per_person"], color='skyblue')
plt.xlabel("Average Wage Rate (₹)")
plt.ylabel("State")
plt.title("Average MGNREGA Wage Rate Per Day (State-wise)")
plt.gca().invert_yaxis() # Highest at top
plt.tight_layout()
plt.show()
```

State-wise Total Expenditure

```
# Group by state and sum total expenditure
state_exp = df.groupby("state_name")["total_expenditure"].sum().sort_values(ascending=False)

# Plot
plt.figure(figsize=(12,6))
plt.plot(state_exp.index, state_exp.values, marker='o', linestyle='-', color='#FF7043')
plt.title("State-wise Total Expenditure under MGNREGA")
plt.xlabel("State")
plt.ylabel("Total Expenditure (₹)")
plt.xticks(rotation=90)
plt.grid(True)
plt.tight_layout()
plt.show()
```

Person-days generated (Top 10)

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_excel("C:/Users/Admin/Downloads/MGNREGA_dataset_AtAGlance18.xlsx")
df.columns = df.columns.str.strip().str.lower()

# Convert numeric safely
df['persondays_of_central_liability_so_far'] = pd.to_numeric(
    df['persondays_of_central_liability_so_far'], errors='coerce'
)

top10 = df.groupby('state_name')['persondays_of_central_liability_so_far'].sum().nlargest(10)

plt.figure(figsize=(10, 6))
bars = plt.bar(top10.index, top10.values, color='yellow', edgecolor='black')

for bar in bars:
    plt.text(bar.get_x() + bar.get_width()/2, bar.get_height(),
             f'{bar.get_height():,}', ha='center', va='bottom', fontsize=9)

plt.title("Top 10 States by Persondays of Central Liability (MGNREGA)", fontsize=14, weight='bold')
plt.xlabel("State", fontsize=12)
plt.ylabel("Persondays (count)", fontsize=12)
plt.xticks(rotation=45, ha='right')
plt.grid(axis='y', linestyle='--', alpha=0.7)

plt.tight_layout()
plt.show()
```